

Unit 8, Lesson 1: Solving Equations

Benchmark

MA.8.AR.2.1 Solve multi-step linear equations in one variable, with rational number coefficients. Include equations with variables on both sides.

Learning Target

- ☐ I can solve multi-step linear equations from a mathematical context.

8.1.1 Warm-Up: Problem Solving

Employers seek out individuals who can collaborate with others to solve problems.

- 1. What are some of the advantages of working with others when problem solving?
- 2. Think of a situation outside of school when you exchanged ideas with someone else to solve a problem. Summarize the steps you took to solve the problem and arrive at the final solution.

8.1.2 Collaboration: Solving Equations

- 1. Da’Liyah and Gerakl both started to solve the equation $3(x - 7) = 18$ for x . Their first steps are shown in green, followed by the resulting equation.

Da’Liyah	Gerald
$\frac{3(x - 7)}{3} = \frac{18}{3}$ $x - 7 = 6$	$3(x - 7) = 18$ $3(x) - 3(7) = 18$ $3x - 21 = 18$

- a. Discuss with your partner what Da’Liyah did differently from Gerakl.
- b. Finish solving the equation for both Da’Liyah and Gerakl, continuing from their first steps.
- c. Verify your answers with your partner and if necessary, resolve any differences.
- d. What do you notice about Da’Liyah’s and Gerald’s solutions to the equation?

2. Two equations are shown in the table.
- a. For each equation, solve it using either Da’Liyah’s or Gerald’s strategy.

Equation	Da’Liyah’s Strategy	Gerald’s Strategy
$-3 = -10(b - 1)$		
$2000(k - 2) = 6000$		

- b. Discuss with your partner why they chose to utilize the specific strategy for each equation. Summarize their reasoning.

8.1.3 Guided Instruction: Like Terms

A **term** is a single number or variable, or numbers and variables multiplied together. For example, the expression $0.5 - 4y$ has two terms.

The equation below includes four terms.

$$a - 21 + 3a = 18$$

- a is a term with a single variable.
- -21 and 18 are each terms with a single number, often referred to as constant terms.
- $3a$ is a term with a number and a variable multiplied together.

When solving equations, only **like terms** can be added or subtracted together. In the equation above, a and $3a$ are like terms and -21 and 18 are like terms.

Like terms are terms with the exact same variable(s) raised to the same exponent.

1. Draw lines to match the like terms in each column.

- | | |
|------------|-------------|
| A. $10p$ | I. -1 |
| B. $12m^2$ | II. $13m^2$ |
| C. f | III. $-3p$ |
| D. -64 | IV. $7f$ |

8.1.4 Collaboration: Error Analysis

Jayla was given the equation $15 = 3c + 6 + 6c$ to solve.

She combined the terms $3c$ and $6c$ by adding them together to create the equation $15 = 9c + 6$.

1. Complete the statement.

Jayla

☐ is

☐ is not

 correct to combine the terms $3c$ and $6c$ because they are

☐ like terms.

☐ not like terms.

Given the same equation, Kenneth combined the terms $3c$, 6 , and $6c$ by adding them together to create the equation $15 = 15c$.

2. Complete the statement.

Kenneth is

☐ correct

☐ incorrect

 to combine the terms $3c$, 6 , and $6c$ because they are

☐ like terms.

☐ not like terms.

3. Discuss with your partner whether or not the equation can be rewritten by adding 15 and 6 together.

8.1.5 Guided Practice: Solving Equations

1. Consider the equation $12 = 4(1 + q - 2)$.
- a. Explain the first step you would take to solve the equation for q .
- b. Compare your first step with your partner. If you chose different first steps, explain why both are correct. If you chose the same first step, work together to describe another first step that could have been taken.
- c. Solve the equation for q .
- d. Use substitution to verify the solution.

2. Three equations are given.

a. Solve each equation on your own using the space provided to show your work.

$12 = 12(-2 - z + 5)$	$2w + 9 - 5w = -3$	$13 = g + 2(g - 4)$

b. Compare your work and solutions with your partner. Resolve any differences in solutions, if necessary.

8.1.6 Wrap-Up: Error Analysis

1. Hubert solved the equation $-8 = 4(2 - h + 1) + 4 - 4h$.
- His work is shown.

$$\frac{-8}{4} = \frac{4(2-h+1)+4-4h}{4}$$
$$-2 = 2-h+1+4-4h$$
$$-2 = 7-5h$$
$$-9 = -5h$$
$$1.8 = h$$

- a. Explain the mistake Hubert made when solving the equation.
- b. Write a short note to Hubert with a tip for how to avoid this error in the future so he can learn from his mistake.

